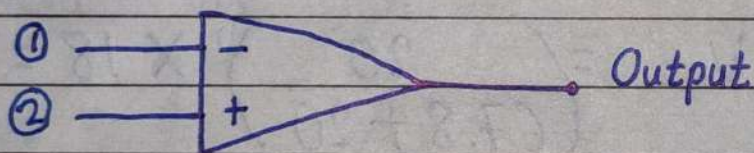


MODULE - III

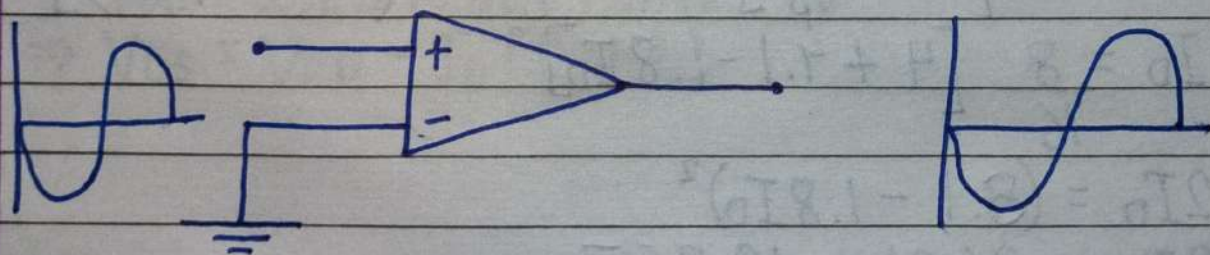
Operational Amplifier & Digital Electronics

Operational Amplifier (Op-Amp)

- It is a very high gain differential amplifier.
- It has very high input impedance and very low output impedance.
- Symbol:

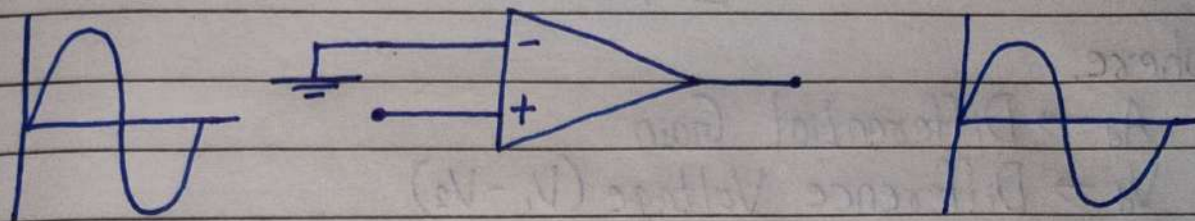


- There are two input terminals: Inverting & Non-Inverting Input Terminals.
- Inverting Input Terminal:
Terminal ① is acting as an inverting input terminal.

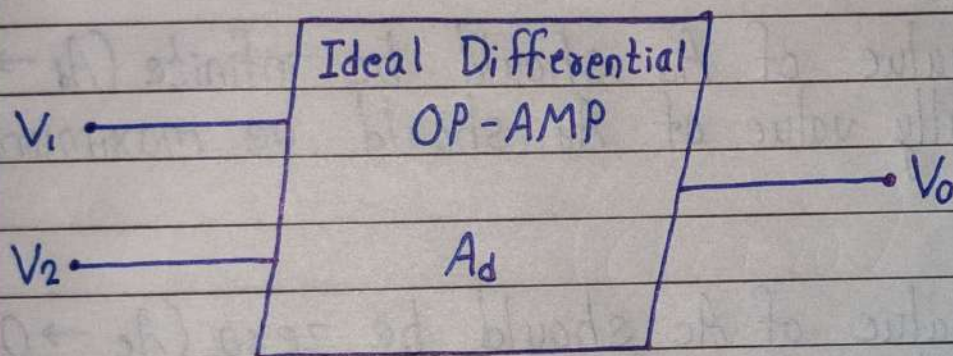


→ Non-Inverting Input Terminal:

Terminal ② is acting as a non-inverting input terminal.



Ideal Differential OP-AMP



$$V_0 \propto (V_1 - V_2)$$

$$\Rightarrow \boxed{V_0 = A_d V_d} \quad [V_d = V_1 - V_2]$$

Common Mode Voltage Gain

We know that for differential amplifier, output is,

$$V_0 = A_d V_d = A_d (V_1 - V_2)$$

If $V_1 = V_2 = V$, then output should be zero but this is not the case and output is not zero, because above amplifier not only amplifies V_d with gain of A_d but also amplifies common voltage V_c with

a gain of A_c .

$$V_o = A_d V_d + A_c V_c$$

Undesired / Unwanted

where,

$A_d \rightarrow$ Differential Gain

$V_d \rightarrow$ Difference Voltage ($V_1 - V_2$)

$A_c \rightarrow$ Common Mode Voltage Gain

$V_c \rightarrow$ Common Voltage ($\frac{V_1 + V_2}{2}$)

$\rightarrow$ The ideal value of A_d should be infinite ($A_d \rightarrow \infty$) but practically value of A_d should be maximum as possible.

$\rightarrow$ The ideal value of A_c should be zero ($A_c \rightarrow 0$) but practically value of A_c should be as low as possible.

Common Mode Rejection Ratio (CMRR)

The ability of the amplifier to reject common voltage V_c is known as CMRR and is given by:-

$$CMRR = \frac{A_d}{A_c}$$

The ideal value of CMRR should be infinite but practically value of CMRR should be as maximum as possible.

Numericals

Q) For an OP-AMP, A_d is 5×10^5 and $A_c = 50$, then calculate CMRR?

Solⁿ:

$$CMRR = \frac{A_d}{A_c} = \frac{5 \times 10^5}{50} = 10^4.$$

Q) Find the output of OP-AMP, when $V_1 = 150 \mu V$, $V_2 = 140 \mu V$, $A_d = 4000$ & CMRR is 100?

Solⁿ:

$$V_1 = 150 \mu V$$

$$V_2 = 140 \mu V$$

$$V_d = V_1 - V_2 = 10 \mu V$$

$$V_c = \frac{V_1 + V_2}{2} = 145 \mu V$$

$$A_d = 4000$$

$$CMRR = 100$$

Now,

$$\frac{1}{A_c} = \frac{CMRR}{A_d} \Rightarrow A_c = \frac{A_d}{CMRR} = \frac{4000}{100} = 40$$

$$\therefore V_o = A_d V_d + A_c V_c = 45,800 \mu V.$$

Imp. # Characteristics of Ideal OP-AMP

- Infinite bandwidth.
- Infinite input impedance.
- Zero output impedance.
- Infinite slew rate.
- Infinite CMRR.

Concept of Virtual Ground

For an ideal OP-AMP, we know that,

$$V_o = A_d V_d$$

$$\Rightarrow V_d = \frac{V_o}{A_d}$$

$$\Rightarrow V_d = 0 \quad [\because A_d \rightarrow \infty \text{ (for ideal)}]$$

$$\Rightarrow V_1 - V_2 = 0$$

$$\Rightarrow \boxed{V_1 = V_2}$$

OP-AMP Parameters

Input Offset Voltage:

It is voltage that should be applied between the input terminals of the correct polarity & magnitude to make the output zero.

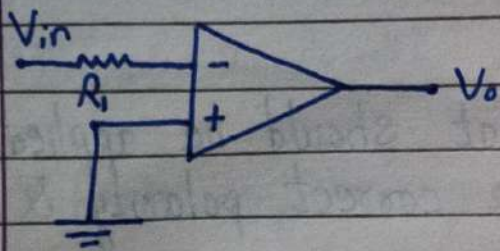
Slew Rate:

It is the maximum rate of change of the output voltage w.r.to time.

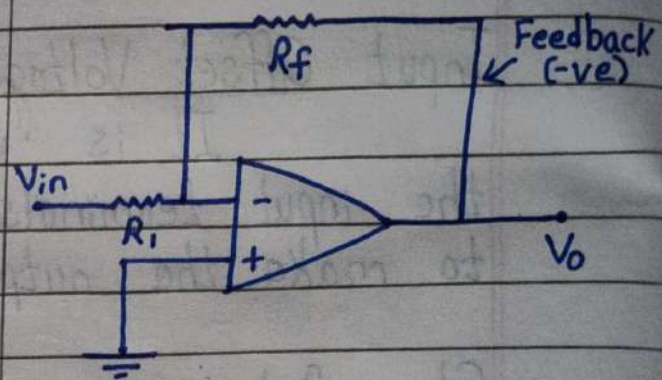
$$\boxed{\text{Slew Rate} = \left[\frac{dV_o}{dt} \right]_{\text{max}}}$$

#

Open Loop

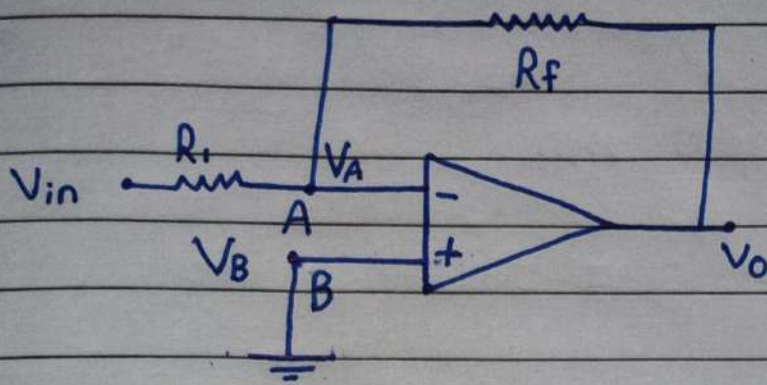


Closed Loop



- | | | | |
|------|---------------------------------|------|---------------------------------|
| i. | Decreases the bandwidth. | i. | Increases the bandwidth. |
| ii. | Decreases the input impedance. | ii. | Increases the input impedance. |
| iii. | Increases the output impedance. | iii. | Decreases the output impedance. |
| iv. | Decreases the stability. | iv. | Increases the stability. |
| v. | High/Uncontrolled gain. | v. | Reduces the voltage gain. |

Ideal Inverting OP-AMP



$$V_A = V_B = 0 \quad [\text{From Concept of Virtual Ground}]$$

Applying KCL at A,

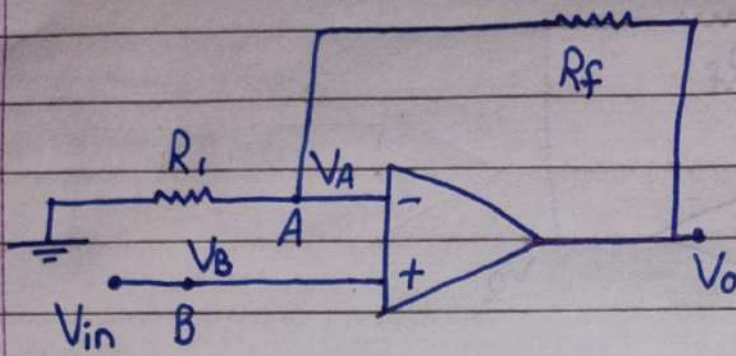
$$\frac{V_A - V_{in}}{R_i} + \frac{V_A - V_o}{R_f} = 0$$

$$\Rightarrow \frac{-V_{in}}{R_i} - \frac{V_o}{R_f} = 0$$

$$\Rightarrow \boxed{V_o = \left(-\frac{R_f}{R_i} \right) V_{in}}$$

$$\text{Voltage Gain} = \frac{V_o}{V_{in}} = -\frac{R_f}{R_i}$$

Ideal Non-Inverting OP-AMP



$$V_A - V_B = 0 \quad [\text{Virtual Ground Concept}]$$

$$\Rightarrow V_A = V_B = V_{in}$$

Applying KCL at A,

$$\frac{V_A - 0}{R_i} + \frac{V_A - V_o}{R_f} = 0$$

$$\Rightarrow \frac{V_{in}}{R_i} + \frac{V_{in} - V_o}{R_f} = 0$$

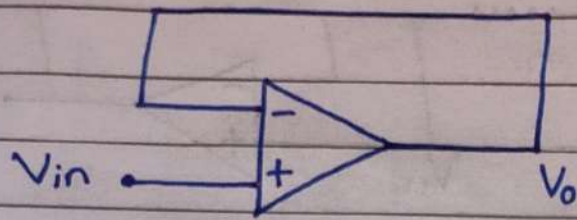
$$\Rightarrow \frac{V_o}{R_f} = \frac{V_{in}}{R_f} + \frac{V_{in}}{R_i}$$

$$\Rightarrow \frac{V_o}{R_f} = V_{in} \left(\frac{1}{R_i} + \frac{1}{R_f} \right)$$

$$\Rightarrow \frac{V_o}{R_f} = \frac{V_{in}}{R_i R_f} (R_f + R_i)$$

$$\Rightarrow V_o = \left(1 + \frac{R_f}{R_i} \right) V_{in}$$

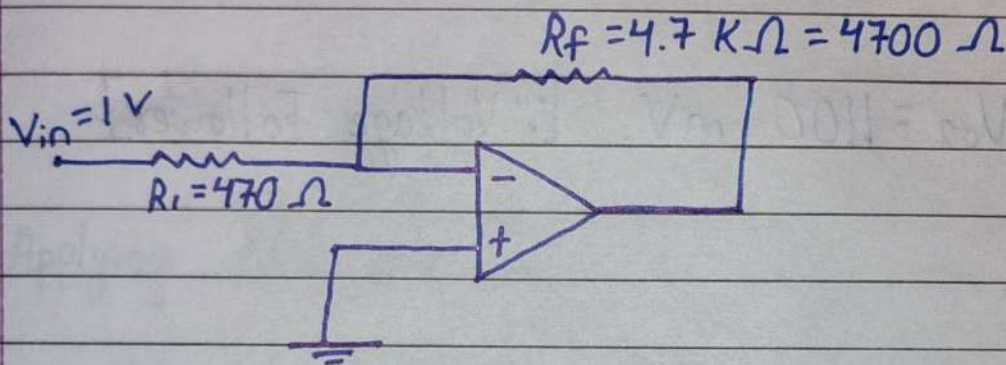
Voltage Follower or Unity Gain Amplifier



$$V_o = V_{in}$$

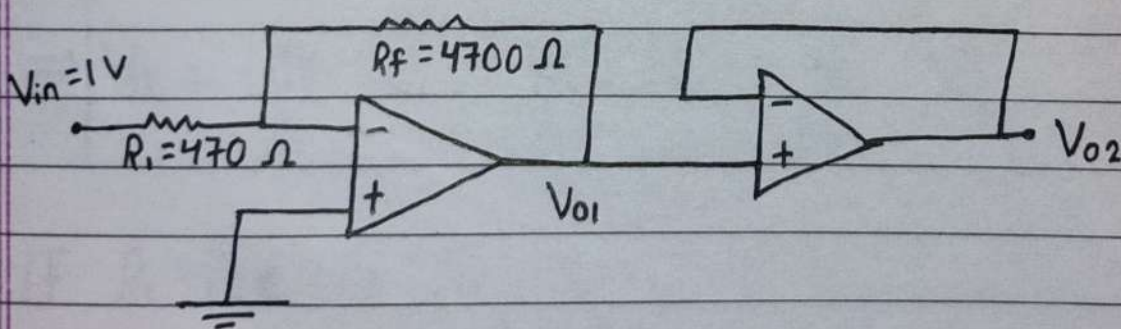
Numericals

Q) For the following circuit, find output voltage V_o ?



$$V_o = \left(\frac{-R_f}{R_i} \right) V_{in} = \left(\frac{-4700}{470} \right) \times 1 = -10 \text{ V.}$$

Q) For the circuit, find V_{o1} & V_{o2} ?

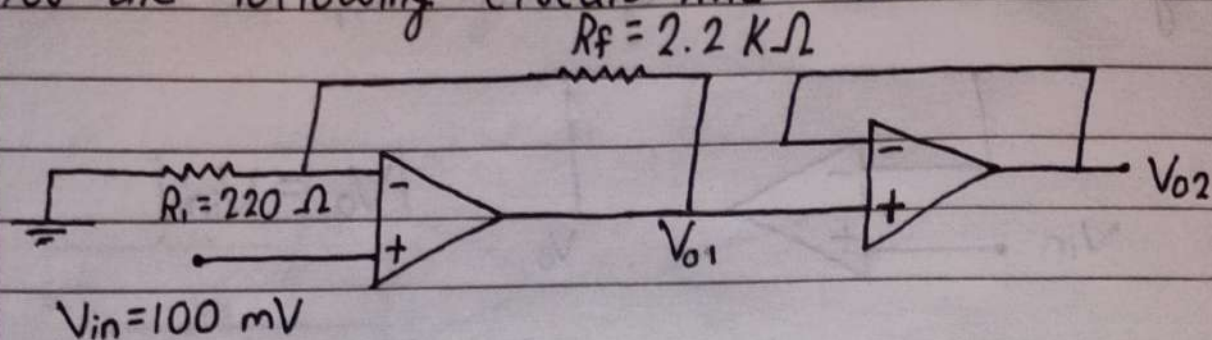


Solⁿ:

$$V_{o1} = \left(\frac{-R_f}{R_i} \right) V_{in} = -10 \text{ V}$$

$$\therefore V_{o2} = V_{o1} = -10 \text{ V} \quad [\because \text{Voltage Follower}]$$

Q) For the following circuit find V_{o1} & V_{o2} ?



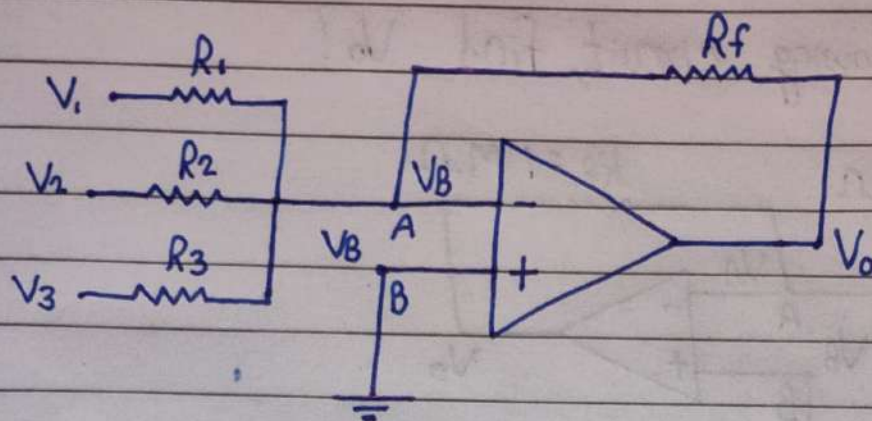
Solⁿ:

$$V_{o1} = \left(1 + \frac{R_f}{R_1}\right) V_{in} = \left(1 + \frac{2200}{220}\right) \times 100 = 1100 \text{ mV}$$

Also,

$$V_{o1} = V_{o2} = 1100 \text{ mV. } [\because \text{Voltage Follower}]$$

V.Imp # Inverting Summing Amplifier



$$V_A - V_B = 0 \quad [\text{Virtual Ground Concept}]$$

$$\Rightarrow V_A = V_B = 0$$

Applying KCL at A,

$$\frac{V_A - V_1}{R_1} + \frac{V_A - V_2}{R_2} + \frac{V_A - V_3}{R_3} + \frac{V_A - V_0}{R_f} = 0$$

$$\Rightarrow \frac{-V_1}{R_1} - \frac{V_2}{R_2} - \frac{V_3}{R_3} - \frac{V_0}{R_f} = 0$$

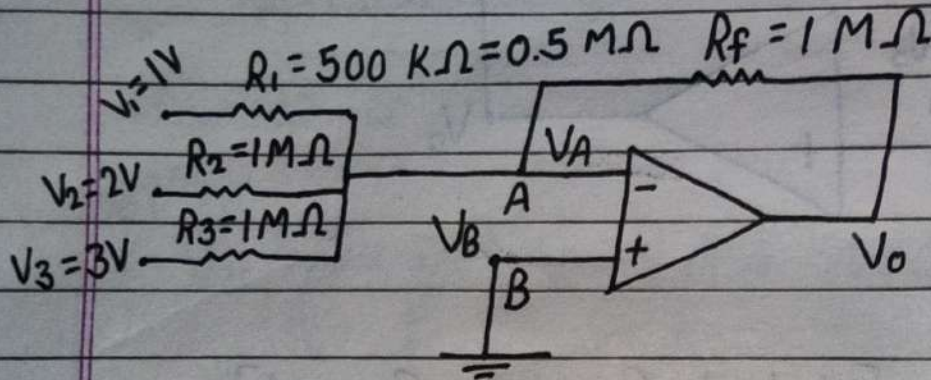
$$\Rightarrow V_0 = -\frac{R_f}{R_1} \cdot V_1 - \frac{R_f}{R_2} \cdot V_2 - \frac{R_f}{R_3} \cdot V_3$$

$$\text{If } R_1 = R_2 = R_3 = R_f = R,$$
$$V_0 = -(V_1 + V_2 + V_3)$$

$$\text{For Average, } R_f = R \text{ \& } R_1 = R_2 = R_3 = 3R,$$
$$V_0 = -\left(\frac{V_1 + V_2 + V_3}{3}\right)$$

Numericals

Q) For the following circuit, find V_o ?



Solⁿ:

$$V_o = -\frac{R_f}{R_1} \cdot V_1 - \frac{R_f}{R_2} \cdot V_2 - \frac{R_f}{R_3} \cdot V_3$$

$$\Rightarrow V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

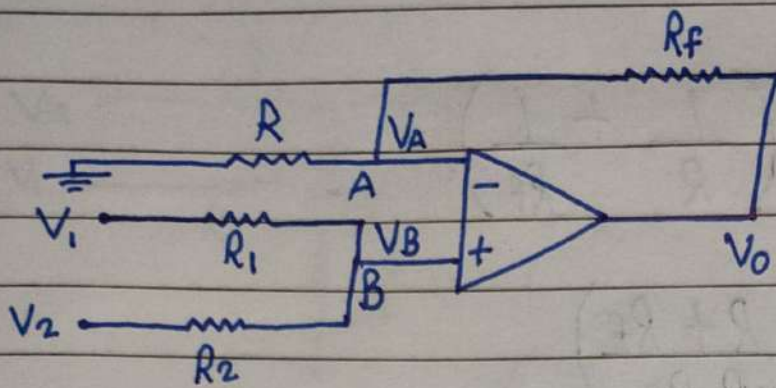
$$\Rightarrow V_o = -1 \left(\frac{1}{0.5} + \frac{2}{1} + \frac{3}{1} \right)$$

$$\Rightarrow V_o = -1 \times (2 + 2 + 3)$$

$$\Rightarrow V_o = -1 \times 7$$

$$\Rightarrow V_o = -7\text{ V.}$$

Non-Inverting Summing Amplifier



$$V_A = V_B \quad [\text{From Virtual Ground Concept}]$$

Applying KCL at B,

$$\frac{V_B - V_1}{R_1} + \frac{V_B - V_2}{R_2} = 0$$

$$\Rightarrow (V_B - V_1)R_2 + (V_B - V_2)R_1 = 0$$

$$\Rightarrow V_B(R_1 + R_2) - V_1R_2 - V_2R_1 = 0$$

$$\Rightarrow V_B = \frac{V_1R_2 + V_2R_1}{R_1 + R_2}$$

$$\text{i.e. } V_A = \frac{V_1R_2 + V_2R_1}{R_1 + R_2}$$

Applying KCL at A,

$$\frac{V_A - 0}{R} + \frac{V_A - V_0}{R_f} = 0$$

$$\Rightarrow \frac{V_A}{R} + \frac{V_A}{R_f} - \frac{V_o}{R_f} = 0$$

$$\Rightarrow \frac{V_o}{R_f} = V_A \left(\frac{1}{R} + \frac{1}{R_f} \right)$$

$$\Rightarrow \frac{V_o}{R_f} = V_A \left(\frac{R + R_f}{R \cdot R_f} \right)$$

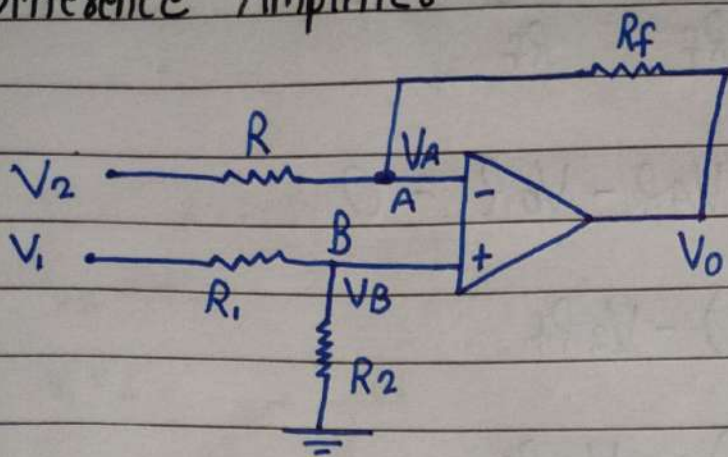
$$\Rightarrow V_o = V_A \left(\frac{R + R_f}{R} \right)$$

$$\Rightarrow V_o = \frac{(V_1 R_2 + V_2 R_1)(R + R_f)}{R(R_1 + R_2)}$$

If $R_1 = R_2 = R_f = R$,

$$V_o = V_1 + V_2.$$

Difference Amplifier



$$V_A = V_B \quad [\text{From Virtual Ground Concept}]$$

Applying KCL at B,

$$\frac{V_B - V_1}{R_1} + \frac{V_B - 0}{R_2} = 0$$

$$\Rightarrow (V_B - V_1)R_2 + V_B R_1 = 0$$

$$\Rightarrow V_B (R_1 + R_2) - V_1 R_2 = 0$$

$$\Rightarrow V_B = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_1$$

$$\text{i.e. } V_A = \left(\frac{R_2}{R_1 + R_2} \right) \cdot V_1$$

Applying KCL at A,

$$\frac{V_A - V_2}{R} + \frac{V_A - V_0}{R_f} = 0$$

$$\Rightarrow \frac{V_A}{R} - \frac{V_2}{R} + \frac{V_A}{R_f} - \frac{V_0}{R_f} = 0$$

$$\Rightarrow V_A R_f - V_2 R_f + V_A R - V_0 R = 0$$

$$\Rightarrow V_0 R = V_A (R + R_f) - V_2 R_f$$

$$\Rightarrow V_0 = V_A \left(\frac{R + R_f}{R} \right) - \frac{V_2 R_f}{R}$$

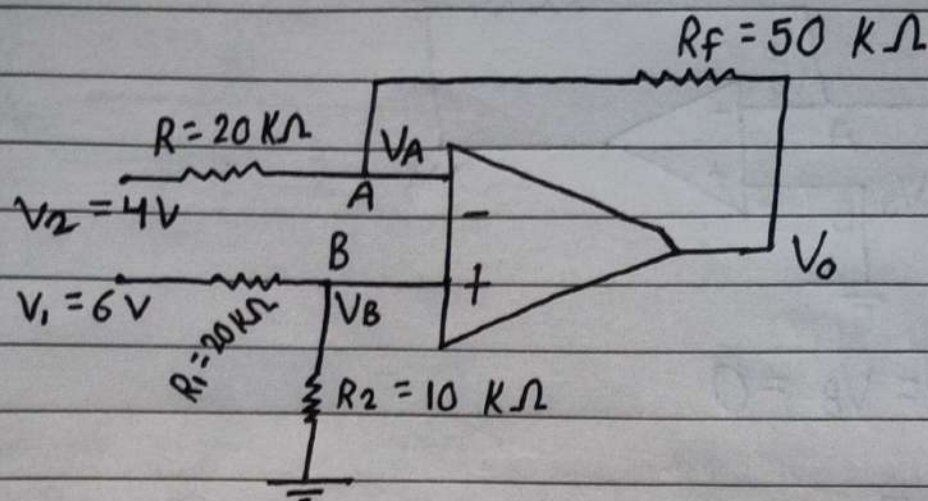
$$\Rightarrow V_0 = \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{R + R_f}{R} \right) V_1 - V_2 \cdot \frac{R_f}{R}$$

$$\Rightarrow V_0 = \frac{R_2 (R + R_f)}{R (R_1 + R_2)} \cdot V_1 - V_2 \cdot \frac{R_f}{R}$$

$$\text{If } R_1 = R_2 = R_f = R,$$

$$V_0 = V_1 - V_2$$

Q) For the following circuit, find output voltage?



Solⁿ:

Using Formula,

$$V_o = \frac{R_2(R+R_f)}{R(R_1+R_2)} \cdot V_1 - \frac{R_f}{R} \cdot V_2 = -3 \text{ V.}$$

OR

Applying KCL at B,

$$\frac{V_B - 6}{20} + \frac{V_B - 0}{10} = 0$$

$$\Rightarrow V_B - 6 + 2V_B = 0$$

$$\Rightarrow V_B = 2 \text{ V}$$

$$\text{i.e. } V_A = V_B = 2 \text{ V}$$

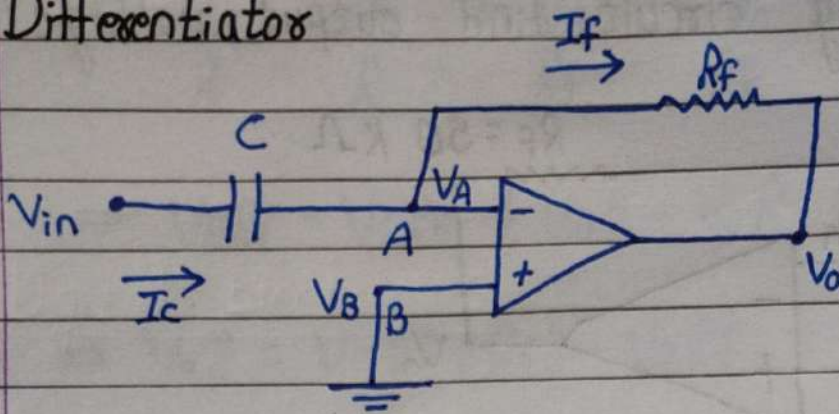
Applying KCL at A,

$$\frac{2 - 4}{20} + \frac{2 - V_o}{50} = 0$$

$$\Rightarrow \frac{-1}{10} + \frac{2 - V_o}{50} = 0$$

$$\Rightarrow V_o = -3 \text{ V.}$$

Differentiator



$$V_A = V_B = 0$$

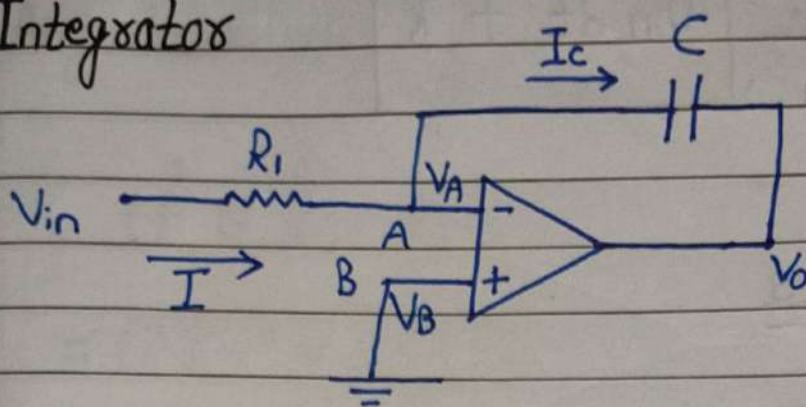
$$\& I_c = I_f$$

$$\Rightarrow C \frac{d}{dt} (V_{in} - V_A) = \frac{V_A - V_o}{R_f}$$

$$\Rightarrow C \frac{d}{dt} V_{in} = -\frac{V_o}{R_f}$$

$$\Rightarrow V_o = -C R_f \frac{dV_{in}}{dt}$$

Integrator



$$V_A = V_B = 0$$

$$\& I = I_c$$

$$\Rightarrow \frac{V_{in} - V_A}{R_i} = C \cdot \frac{d}{dt} (V_A - V_o)$$

$$\Rightarrow \frac{V_{in}}{R_i} = C \frac{d}{dt} (-V_o)$$

$$\Rightarrow \frac{V_{in}}{R_i} = -C \frac{d}{dt} V_o$$

$$\Rightarrow -\frac{1}{CR_i} \cdot V_{in} dt = dV_o$$

$$\Rightarrow dV_o = -\frac{1}{CR_i} V_{in} dt$$

Integrating both sides,

$$\int dV_o = -\int \frac{1}{CR_i} V_{in} dt$$

$$\Rightarrow V_o = -\frac{1}{CR_1} \int V_{in} dt + K$$

Digital Electronics

Number System

Decimal Base = 10	Binary Base = 2	Octal Base = 8	Hexa-decimal Base = 16
0	0	0	0 to 9
1	1	1	10 - A
2		2	11 - B
3		3	12 - C
4		4	13 - D
5		5	14 - E
6		6	15 - F
7		7	
8			
9			